

Setup Guide: Sumo Ring Push-Out

Time Allocation: 90 Minutes

Challenge Overview

The robot must stay inside a circular ring while locating and pushing a passive target object outside the boundary.

Environment & Arena Setup

- **Arena Construction:** Create a circular arena on a flat, smooth surface (such as a large piece of whiteboard, foam board, or smooth flooring). A standard diameter of 1.5 to 2 meters is recommended for classroom robotics kits.
- **Boundary Marking:** Clearly outline the perimeter of the circle with a high-contrast boundary line (e.g., 2-inch wide black electrical tape on a white background, or vice versa) so that downward-facing color/light sensors can easily detect it.
- **Target Object:** Place a passive, lightweight target object in the center of the ring. The object should be visible to sensors and easy to push, such as a small plastic cup, a weighted soda can, or a 3D-printed block.
- **Testing Zones:** Ensure there is enough surrounding floor space outside the ring boundary for the target object to be safely pushed out without falling off elevated tables, or use a raised ring with a short drop/barrier to catch ejected objects.